



HACK2SKILL

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : NeuralAxis

Team Leader Name : Poornima Menon

Problem Statement : AI-enabled Detection of Exoplanets from Noisy Astronomical Light Curves Objective

Team Members

Team Leader:

Name: Poornima Menon

College: Usha Mittal Institute of Technology

Team Member-1:

Name: Vaishnavi Balodhi

College: Usha Mittal Institute of Technology

Team Member-2:

Name: Divya Sharma

College: Usha Mittal Institute of Technology

Team Member-3:

Name:

College:

Target System Characterization & Follow-up



- Host Star Spectral Classification (K, G, M)
- Estimated Stellar Mass/Radius
- Confirmed Planetary Mass (via Radial Velocity)
- Atmospheric Contamination (Low, Medium, High)
- Optimal Follow-up Instruments (e.g., JWST, Keck)
- Estimated Contamination (Low, Medium)



Extremely Weak Transit Signals

Tiny dips are buried beneath stellar noise. Small planets are challenging to detect.



False Positives Mimic Transits

Eclipsing binaries and instrumental effects resemble planetary transits.



Crowded-Field Contamination

Nearby stars blend signals, confusing source identification.



Manual Vetting Does Not Scale

Large TESS datasets make manual inspection impractical for high-rate discovery.



Detection Alone Is Not Enough

Classification, confidence scoring, and transit parameters are needed.

Target Observation Summary

Data Sector Count

36 (Multi-Sector Campaign)
For deeper analysis

Time Sectors Merts

18.1 SA550
Decimated analysis

Rinching Number

1-minute

Data Sector Count

36 (Multi-Sector Campaign)
For deeper analysis

Light Curve Cadence

2-minute (and 20-sec)

Processing Version

SPOC Data Release 6.2 (v1)



30K+

Target Light Curves



4

Core Scientific Outputs



3+

Major False Positive Sources



1

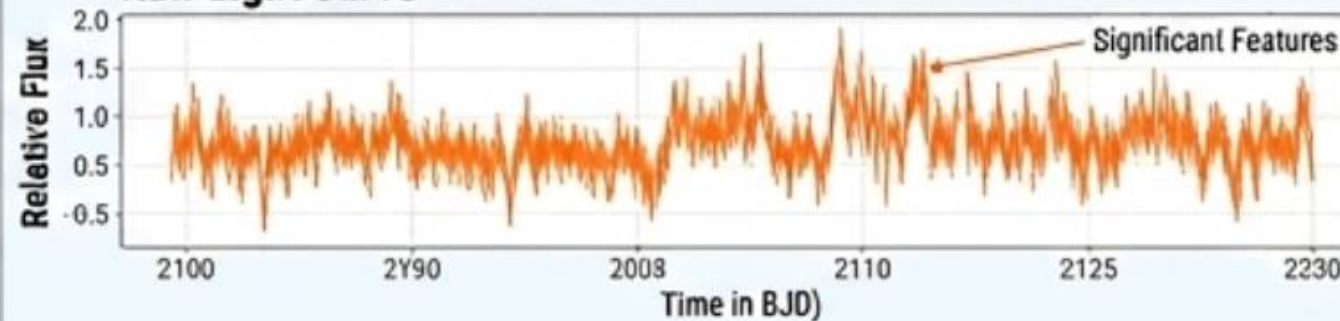
Primary Bottleneck (Vetting)



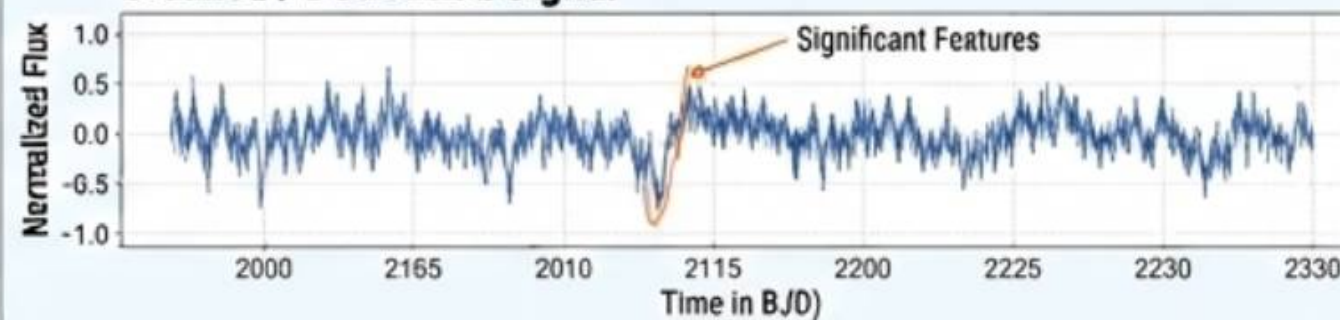
2

Essential Processing Stages (Detection & Vetting)

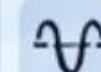
Raw Light Curve



Cleaned / Detrended Signal



AI Vetting Output



Estimated Transit Parameters



Estimated Transit Parameters



Transit Parameter | Sample b_0



Signal Significance (S/N) | High



Transit Impact Parameter | Sample b_0



Signal Significance (S/N) | High

Traditional Analysis

Manual inspection

Separate tools

High false positives

Limited scalability

Basic detection

Proposed AI-Assisted Pipeline

Automated preprocessing

Unified workflow

AI-based classification

Large-scale automated vetting

Ranked, explainable candidates

Sources of Transit-Like Signals

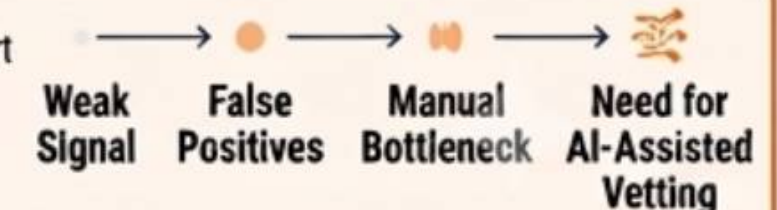


Difficulty

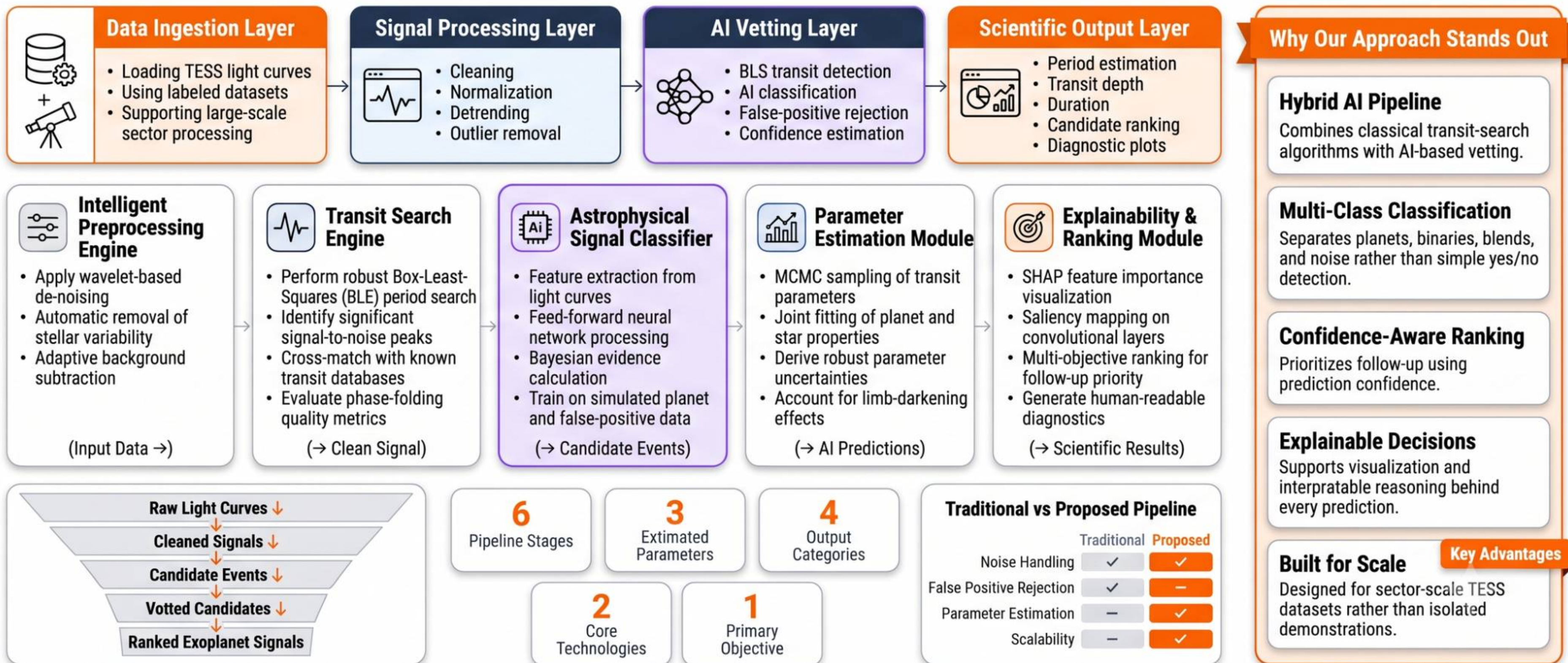


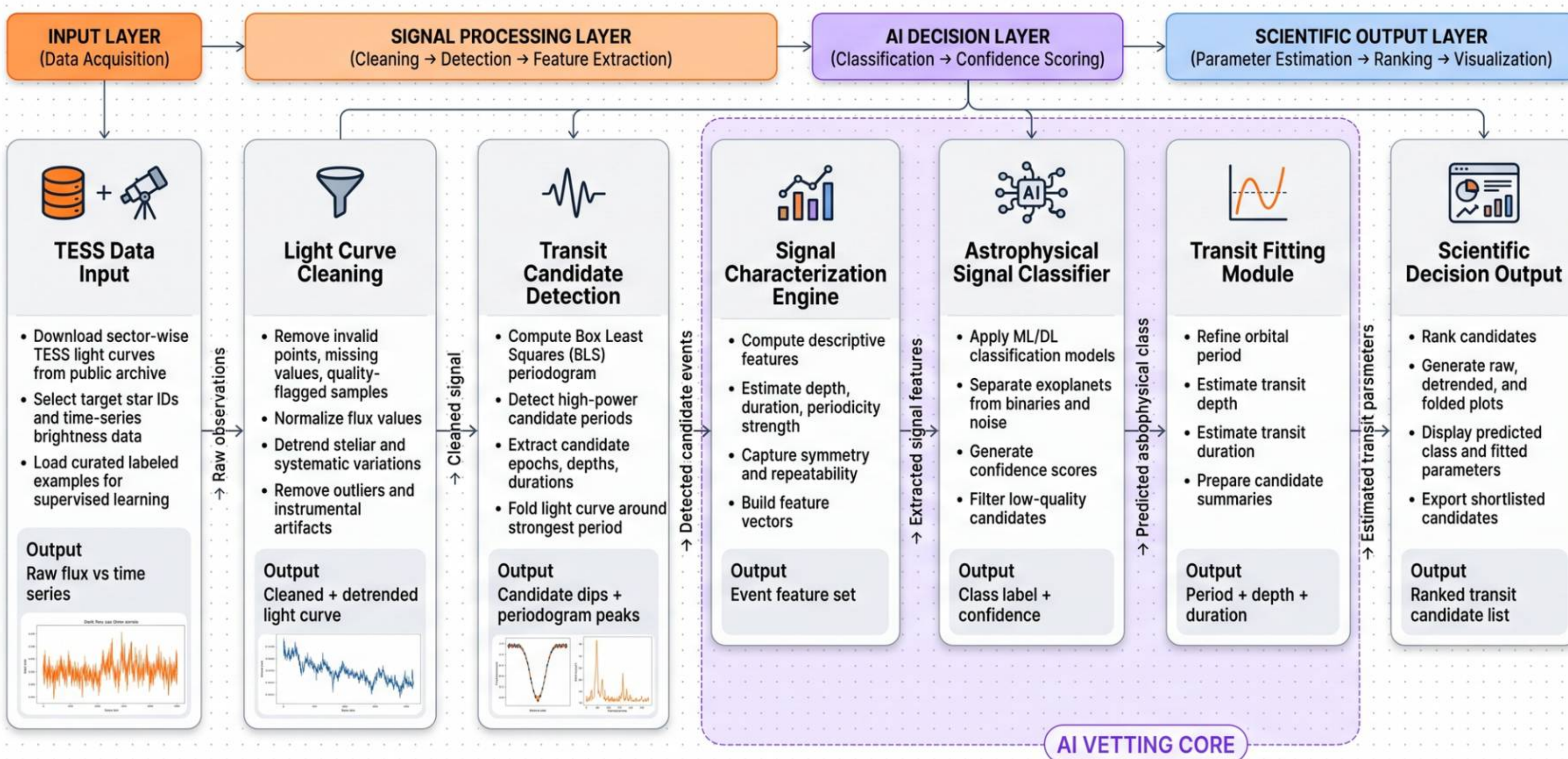
Impact of the Current Gap

- Weak transit events remain undetected
- High false-positive rates increase verification effort
- Manual review slows scientific discovery
- Poor prioritization delays follow-up observations
- Detection-only systems provide incomplete scientific insight



List of features offered by the solution





Why This Workflow Is Effective

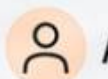
- Separates weak signal discovery from false-positive rejection
- Supports automation and scientific interpretability
- Scales efficiently to massive TESS datasets



Key Decision Points

- Is the dip periodic?
- Is the folded curve transit-like?
- Planet, binary, or contamination?
- Confidence high enough for follow-up?

Practical Research Workflow



Astronomer



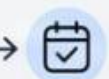
Select TESS Sector



Run Pipeline



Inspect Ranked Candidates



Validate Best Transit Events



Target Observation Summary

TIC ID: **TIC-28491271**
Sector: **28**
Cadence: **2-minute**
Observation Span: **27 Days**
Quality Score: **Good**
Contamination Risk: **Low**

Detected Candidate Events

3

Top BLS Peak

4.26 d

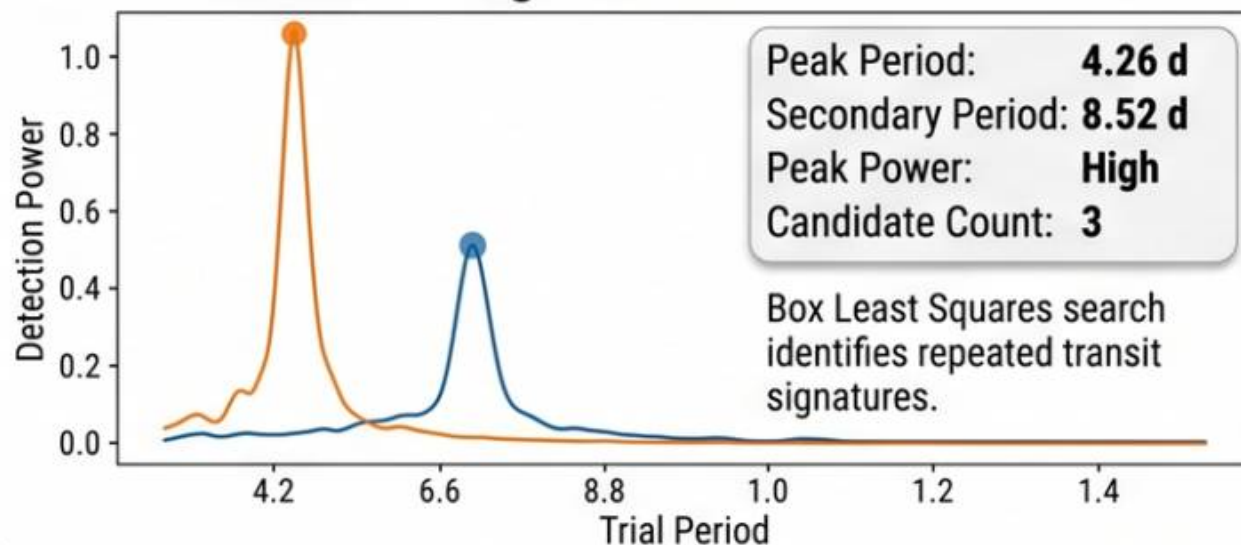
Best Confidence

91%

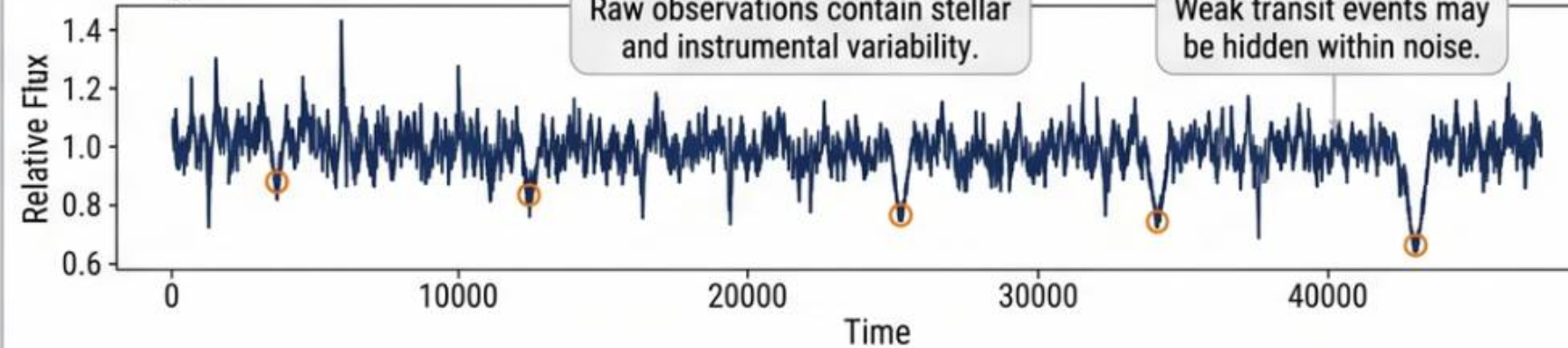
Transit Ranking

1 / 3

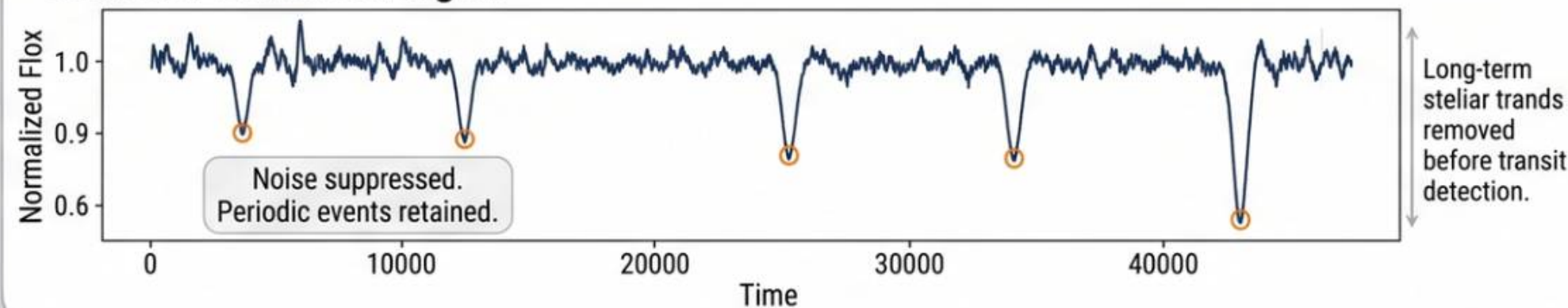
Transit Search Periodogram



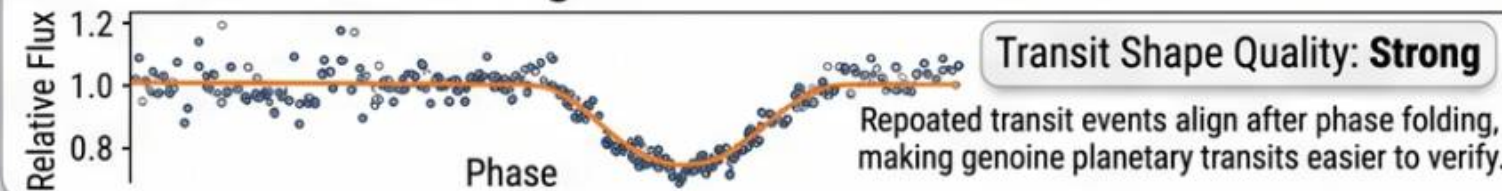
Raw Light Curve



Cleaned / Detrended Signal



Phase-Folded Transit Signal



Rank	Candidate	Predicted Class	Confidence	Priority
1	TIC-01	Probable Transit	0.91	High
2	TIC-02	Possible Binary	0.67	Medium
3	TIC-03	Noise / Blend	0.31	Low

AI Vetting Output

Predicted Class: **Probable Exoplanet Transit**
Confidence: **91%**

Probable Exoplanet **91%**
Eclipsing Binary **6%**
Blend / Contamination **3%**

Multi-class classification reduces false positives and prioritizes high-quality candidates.

Estimated Transit Parameters

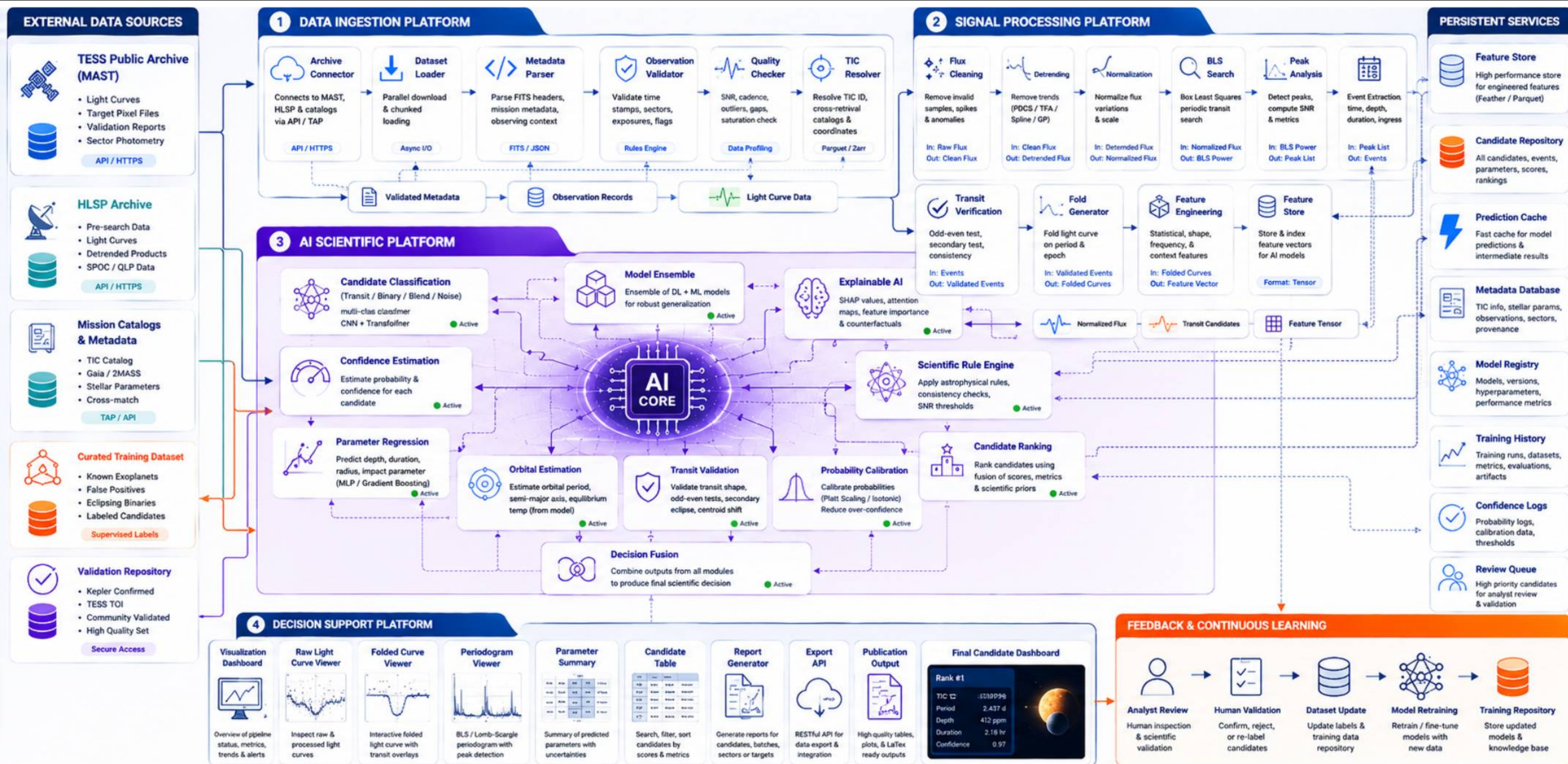
Orbital Period **4.26 days**
Transit Depth **0.8%**
Transit Duration **2.1 hr**
Epoch **Sample T_0**
Signal Significance **High**

Explainable AI

Why the Model Flagged this Candidate

- Periodic dip repeats consistently.
- Folded light curve shows a compact transit signature.
- Signal remains visible after preprocessing.
- Competing false-positive probabilities are significantly lower.

Architecture diagram of the proposed solution



TECHNOLOGIES TO BE USED IN THE SOLUTION

A modern, open-source technology stack for astronomical data access, signal processing, machine learning, and scientific insight.



WHY THESE TECHNOLOGIES?

Open Source
Transparent, flexible, and widely adopted by the scientific community.

Interoperable
Seamlessly work together across data, AI, and visualization.

Scalable
Suitable for small experiments to large-scale transit searches.

Research Ready
Built for accuracy, interpretability, and reproducible science.

Community Driven
Backed by active communities and continuous improvements.

— Data Flow — Processing Flow — Storage Flow — Visualization Flow — Deployment Flow — Feedback Loop



BUILT FOR DISCOVERY

This integrated technology stack enables an end-to-end exoplanet detection pipeline—from raw TESS data to ranked, explainable, and scientifically validated exoplanet candidates.

ESTIMATED IMPLEMENTATION COST (OPTIONAL)

Low-Cost, Student-Feasible Prototype with Optional Cloud GPU Scaling

'Project Cost Breakdown'



Public Astronomical Data
TESS light curves are publicly available through the MAST archive.

\$0

FREE



Lightcurve

Open Source Software
All core libraries are open source.

\$0

FREE



Local Development
Prototype development and testing can be performed locally.

\$0

OPTIONAL



Cloud GPU (Optional)
Optional cloud GPUs accelerate model training and experimentation.

\$20–100

OPTIONAL



Large Scale Deployment
Only required for large datasets or production deployment.

\$100+

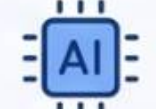
SCALABLE

'Project Feasibility'



Cost Efficiency

- Near-zero prototype cost
- Open-source ecosystem
- No licensing fees



Technical Feasibility

- Python ecosystem
- Public datasets
- Modern ML libraries



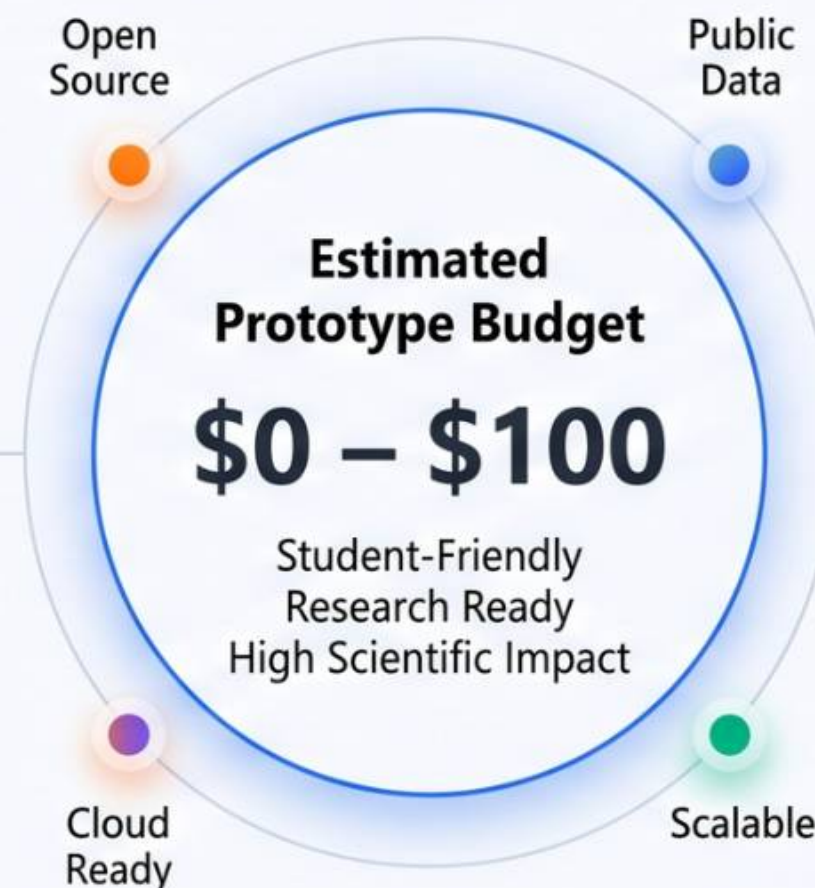
Research Value

- Real astronomical observations
- Explainable AI
- Scientific validation



Future Scalability

- Cloud deployment
- GPU acceleration
- Larger datasets
- Enterprise-ready architecture



The proposed solution is economically feasible for a student-team prototype because it leverages publicly available TESS data, open-source scientific software, and optional pay-per-use cloud resources only when additional computational power is required.



**BHARATIYA
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HACKATHON**

2026

THANK YOU